

Unit 1: Electrostatics

⇒ Electrostatic potential.

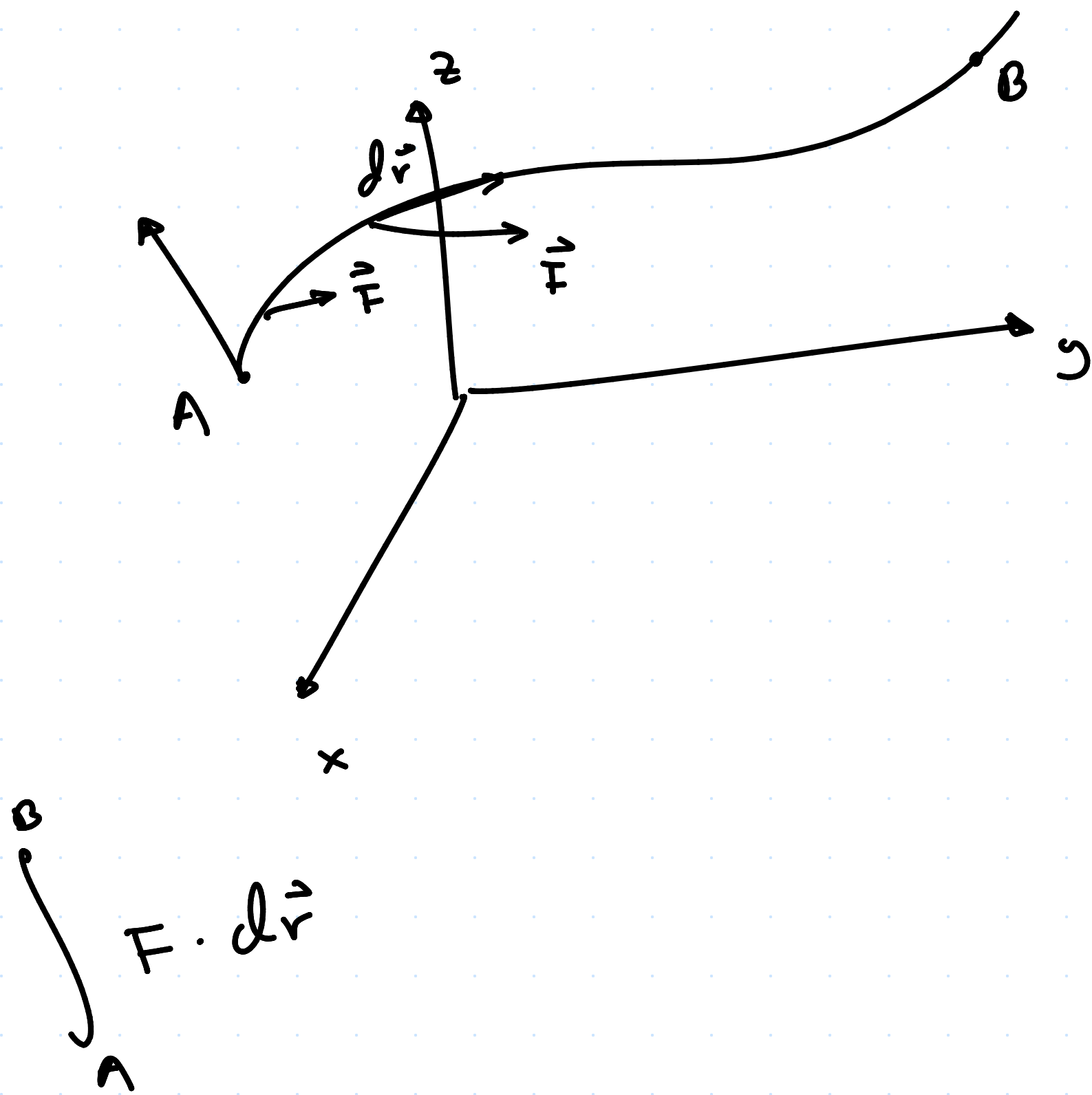
$$U: V \subset \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\vec{E} = -\nabla U$$

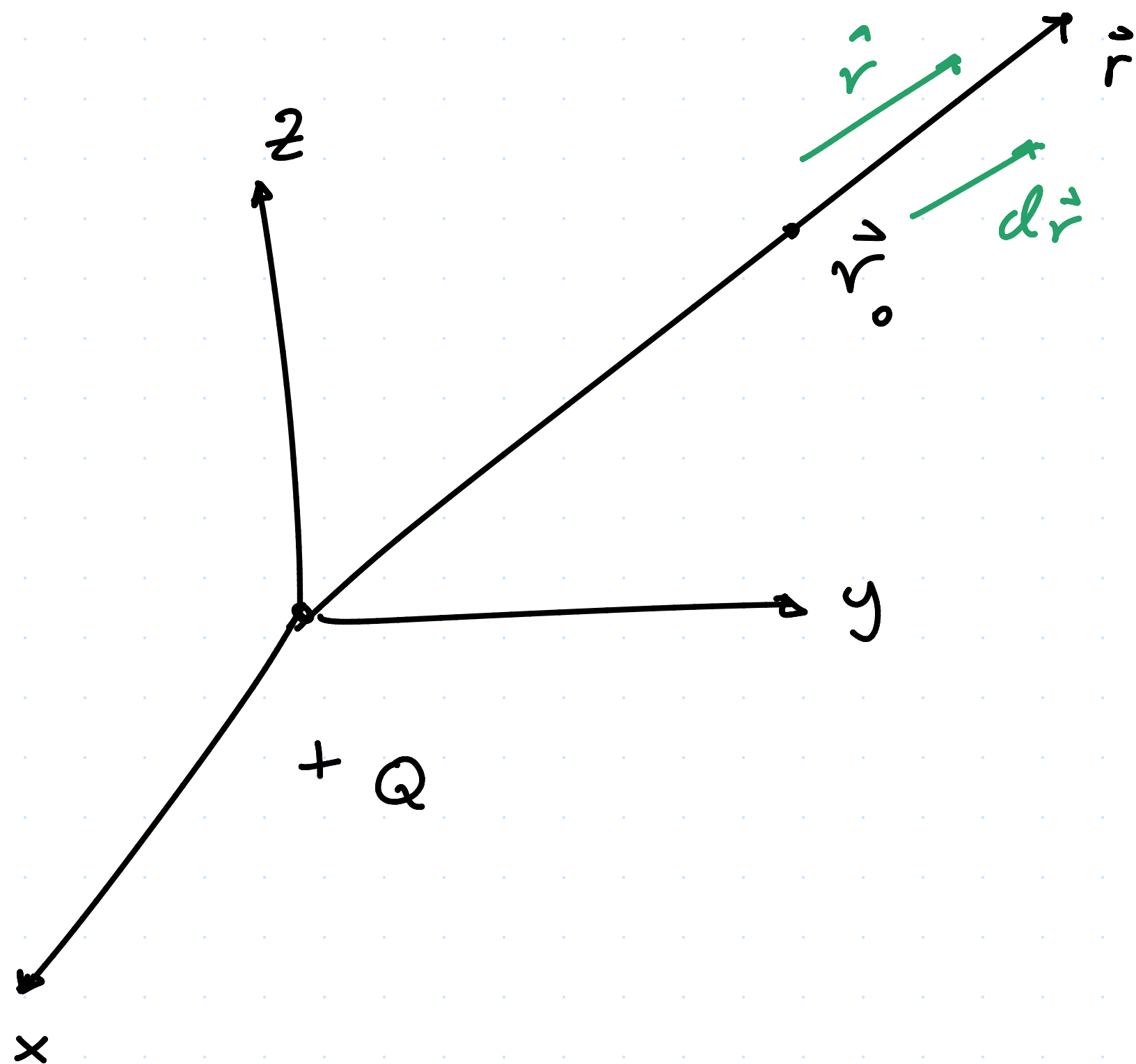
$$\nabla \times \vec{E} = 0$$

scalar  $\mathbb{R}^3 \mapsto \mathbb{R}$  vectorial

Note:



Campo eléctrico



en la misma  
dirección que  $\vec{r}$



$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$

$$\Rightarrow \int_{\vec{r}_0}^{\vec{r}} \vec{E} \cdot d\vec{r} = - \int_{\vec{r}_0}^{\vec{r}} \nabla U \cdot d\vec{r} = - \left[ U(\vec{r}) - U(\vec{r}_0) \right]$$

$$\int_{\vec{r}_0}^{\vec{r}} \vec{E} \cdot d\vec{r} = \frac{1}{4\pi\epsilon_0} \int_{r_0}^r \frac{Q}{r^2} \hat{r} \cdot d\vec{r} = \frac{Q}{4\pi\epsilon_0} \int_{r_0}^r \frac{dr}{r^2}$$

$$= - \frac{Q}{4\pi\epsilon_0} \left( \frac{1}{r} \right) \Big|_{r_0}^r = - \frac{Q}{4\pi\epsilon_0} \left( \frac{1}{r} \right)$$

Usamos una convención

$$U(\vec{r}_0) = 0 \quad \text{for } \vec{r}_0 \rightarrow \infty$$

Significa que estamos  
alejados de la fuente.

Note

$$U = mgh \quad | \quad U = mgy$$

Entonces

$$\cancel{U(\vec{r}_0)} \rightarrow 0 \quad \text{when } r_0 \rightarrow \infty$$

Carga puntual positiva en el origen:

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$



Para el campo eléctrico dado por un punto, superficie y distribuciones de carga volumétrica.

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|\vec{r} - \vec{r}_i|} + \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') d\vec{a}'}{|\vec{r} - \vec{r}'|} + \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}'') d\vec{v}''}{|\vec{r} - \vec{r}''|}$$

$$\rightarrow -\nabla U$$

(a) Point charges

$$\nabla \left( \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|\vec{r} - \vec{r}_i|} \right) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n q_i \nabla \left( \frac{1}{|\vec{r} - \vec{r}_i|} \right)$$

Note!

$$\nabla \left( \frac{1}{|\vec{r} - \vec{r}_i|} \right)_i = \frac{\partial}{\partial x_i} \left[ \frac{1}{\sqrt{(x-x_i)^2 + (y-y_i)^2 + (z-z_i)^2}} \right]$$

↑  
componente i-ésima

Tenemos

$$= \partial_i \left( (x-x_i)^2 + (y-y_i)^2 + (z-z_i)^2 \right)^{-1/2}$$

$$\therefore = -\frac{1}{2} \left( (x-x_i)^2 + (y-y_i)^2 + (z-z_i)^2 \right)^{-3/2} \cdot \left( \sum_{i=1}^3 (x_i - x_{ii})^2 \right)'$$

Delta de  
Kronecker  
↓

$$\hookrightarrow 2 \sum_{j=1}^3 (x_j - x_{ij}) \cdot \delta_{ij}$$



Desarrollado paso a paso

$$\frac{\partial}{\partial x_i} \left( \sum_{j=1}^3 (x_j - x_{1j})^2 \right)$$

$$= \frac{\partial}{\partial x_i} \left[ (x_1 - x_{11})^2 + (x_2 - x_{12})^2 + (x_3 - x_{13})^2 \right]$$

$$\vdots = 2(x_1 - x_{11}) \cdot \frac{\partial}{\partial x_i} (x_1 - x_{11}) + 2(x_2 - x_{12}) \cdot \frac{\partial}{\partial x_i} (x_2 - x_{12})$$

$$+ 2(x_3 - x_{13}) \cdot \frac{\partial}{\partial x_i} (x_3 - x_{13})$$

$$\delta_{ij} \begin{cases} 1 & ; \quad i=j \\ 0 & ; \quad i \neq j \end{cases}$$

$$\rightarrow = 2 \sum_{j=1}^3 (x_j - x_{1j}) \cdot \delta_{ij}$$

cuando  $j=i$  sobrevive  $(x_i - x_{1i})$

Tenemos así:

$$= - \frac{x_i - x_{1i}}{(x_i - x_{1i})^{3/2}} = - \frac{(\vec{r} - \vec{r}_1)_i}{|\vec{r} - \vec{r}_1|^3}$$

Tenemos así:

$$\nabla \left( \frac{1}{|\vec{r} - \vec{r}_1|} \right) = - \frac{(\vec{r} - \vec{r}_1)}{|\vec{r} - \vec{r}_1|^3}$$



Así  $-\nabla U$  es:

$$-\nabla \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|\vec{r}-\vec{r}_i|} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i (\vec{r}-\vec{r}_i)}{|\vec{r}-\vec{r}_i|^3}$$

Electric field due to  
Point charges.

Realizar... b) Surface charge distribution.

Dem.  $-\nabla \left[ \int_S \frac{1}{4\pi\epsilon_0} \frac{\sigma(\vec{r}') d\alpha'}{|\vec{r}-\vec{r}'|} \right] = \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') d\alpha' (\vec{r}-\vec{r}')}{|\vec{r}-\vec{r}'|^3}$

Tenemos...

$$-\nabla \left[ \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') d\alpha'}{|\vec{r}-\vec{r}'|} \right] = - \left[ \frac{1}{4\pi\epsilon_0} \int_S d\alpha' \left( \nabla \left( \frac{\sigma(\vec{r}')}{|\vec{r}-\vec{r}'|} \right) \right) \right]$$

$$\nabla \left( \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|\vec{r}-\vec{r}_i|} \right) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n q_i \nabla \left( \frac{1}{|\vec{r}-\vec{r}_i|} \right) = - \left[ \frac{1}{4\pi\epsilon_0} \int_S d\alpha' \left( \partial_i \frac{\sigma(\vec{r}')}{((x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2)^{3/2}} \right) \right]$$

usamos el resultado anterior para  $\nabla \left( \frac{1}{|\vec{r}-\vec{r}_i|} \right) = - \frac{(\vec{r}-\vec{r}_i)}{|\vec{r}-\vec{r}_i|^3}$

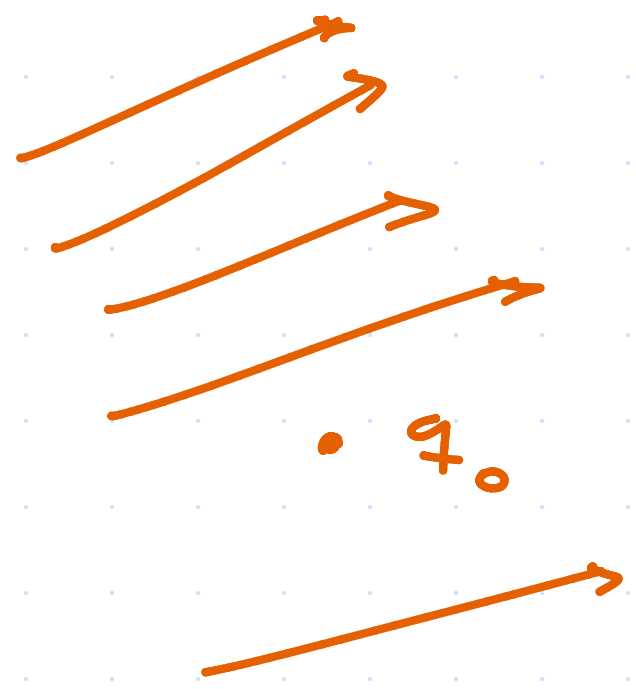
The potential energy can be computed by



$$W = - \int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r} = - \int_{\vec{r}_0}^{\vec{r}} q_0 (-\nabla U) \cdot d\vec{r} \Rightarrow \boxed{W = q_0 U(\vec{r})}$$

$$U = \frac{\text{Joule}}{\text{Coulomb}} = \text{volt (V)}$$

$\vec{F}$



$$\vec{F} = q_0 \vec{E}$$

$$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}_{q_0}}{q_0}$$

Tarea  $\rightarrow$  teoría de bandas electrónicas.